Topic 9 — Thermodynamics

9.1 Internal Energy

Internal energy is the sum of the randomly distributed kinetic energies and potential energies of all the molecules in a system. It is not the same as temperature — internal energy depends on both temperature and the state of the substance.

The **kinetic energy** component is associated with the random translational, rotational, and vibrational motion of molecules. It depends on temperature — higher temperature means greater average kinetic energy.

The **potential energy** component is associated with the intermolecular forces (bonds) between molecules. It depends on the separation of molecules and therefore on the state of matter.

When you heat a substance:

- **If temperature rises:** internal energy increases due to increased kinetic energy of molecules
- **If a change of state occurs (at constant temperature):** internal energy increases due to increased potential energy as molecules move further apart (bonds breaking), but the kinetic energy remains constant

Exam Tip: A common exam question asks why temperature stays constant during a change of state. The answer: all energy input goes into breaking intermolecular bonds (increasing potential energy) rather than increasing kinetic energy. Since temperature is related to the average kinetic energy, it does not change.

9.2 Specific Heat Capacity

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Specific heat capacity (c) is the energy required to raise the temperature of 1 kg of a substance by 1 K (or 1°C), without a change of state.
Unit: J kg^{-1} K^{-1}

$$Q = mc\Delta\theta$$

where Q = energy transferred (J), m = mass (kg), c = specific heat capacity (J kg^{-1} K^{-1}), Δθ = temperature change (K or °C).

Substance	Specific Heat Capacity (J kg^{-1} K^{-1})
Water	4200
Copper	390
Aluminium	900
Lead	130
Air (at constant pressure)	1010
Glass	840

Worked Example: A 2.0 kg aluminium block at 20°C is heated with a 50 W heater for 5 minutes. Calculate the final temperature, assuming no heat loss. (c for aluminium = 900 J kg^{-1} K^{-1})

Energy supplied: Q = Pt = 50 × (5 × 60) = 15 000 J
Q = mcΔθ, so Δθ = Q / (mc) = 15 000 / (2.0 × 900) = 8.3 K
Final temperature = 20 + 8.3 = **28.3°C**

In practice, the actual temperature rise would be less because some heat is lost to the surroundings.

Experimental Determination

A typical experiment uses an electrical heater immersed in a block or liquid:

- Measure mass of substance (m) using a balance
- Record initial temperature (θ_1) using a thermometer
- Heat with known power P for time t, so energy supplied = Pt = VIt
- Record final temperature (θ_2)
- Calculate: $c = Pt / (m \times \Delta\theta)$

Sources of error: heat losses to surroundings (gives a value of c that is too high), uneven heating, thermometer response time. Insulation, stirring, and a snug-fitting heater in a purpose-drilled hole help reduce errors.

Exam Tip: In calculations, remember that energy supplied by the heater may not all go into raising the temperature — some is lost to the surroundings. The experimental value of c will typically be **higher** than the true value because you need to supply more energy than $mc\Delta\theta$ to achieve the observed temperature change.

9.3 Specific Latent Heat

Specific latent heat (L) is the energy required to change the state of 1 kg of a substance without a change in temperature.

Specific latent heat of fusion (L_f): solid → liquid (or liquid → solid)

Specific latent heat of vaporisation (L_v): liquid → gas (or gas → liquid)

Unit: J kg^{-1}

$$Q = mL$$

During a change of state, temperature remains constant because all energy input goes into breaking/forming intermolecular bonds (changing potential energy), not increasing kinetic energy.

L_v is always much larger than L_f for the same substance because vaporisation requires completely separating molecules against intermolecular forces, whereas fusion only requires disrupting the ordered lattice structure.

Substance	L_f (kJ kg^{-1})	L_v (kJ kg^{-1})
Water	334	2260
Lead	23	871
Aluminium	397	10 900

Worked Example: How much energy is needed to convert 0.50 kg of ice at −10°C to steam at 100°C?

Step 1 — Heat ice from −10°C to 0°C: $Q_1 = mc_{ice}\Delta\theta = 0.50 \times 2100 \times 10 = 10\,500$ J

Step 2 — Melt ice at 0°C: $Q_2 = mL_f = 0.50 \times 334\,000 = 167\,000$ J

Step 3 — Heat water from 0°C to 100°C: $Q_3 = mc_{water}\Delta\theta = 0.50 \times 4200 \times 100 = 210\,000$ J

Step 4 — Vaporise water at 100°C: $Q_4 = mL_v = 0.50 \times 2\,260\,000 = 1\,130\,000$ J

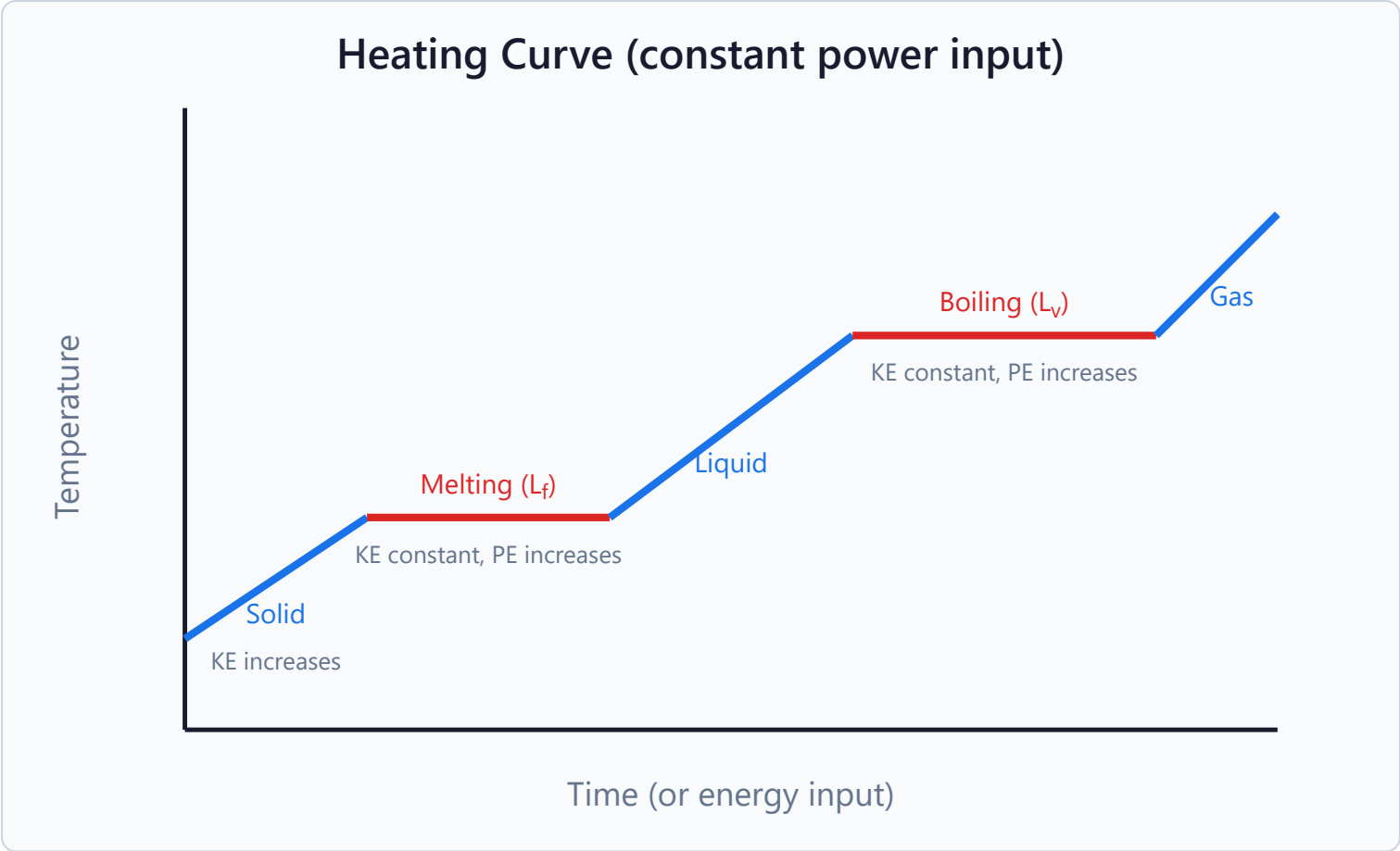
Total = 10 500 + 167 000 + 210 000 + 1 130 000 = **1 517 500 J ≈ 1.5 MJ**

Note that most of the energy (74%) goes into vaporisation alone.

Heating Curves

A heating curve plots temperature against time (or energy) for a substance being heated at constant power:

- **Sloped sections:** Temperature is rising — substance is in a single phase. Gradient depends on specific heat capacity (steeper = lower c or lower mass).
- **Flat sections:** Temperature constant — change of state occurring. Length depends on specific latent heat (longer = higher L or greater mass).



9.4 Gas Laws

Boyle's Law

Boyle's law: For a fixed mass of gas at constant temperature, the pressure is inversely proportional to the volume.
pV = constant (at constant T)

$$p_1V_1 = p_2V_2$$

Microscopic explanation: At constant temperature, reducing the volume means the same number of molecules occupy a smaller space. They hit the walls more frequently (shorter distance between walls), so the force per unit area (pressure) increases.

Charles's Law

Charles's law: For a fixed mass of gas at constant pressure, the volume is directly proportional to the absolute temperature.
V / T = constant (at constant p)

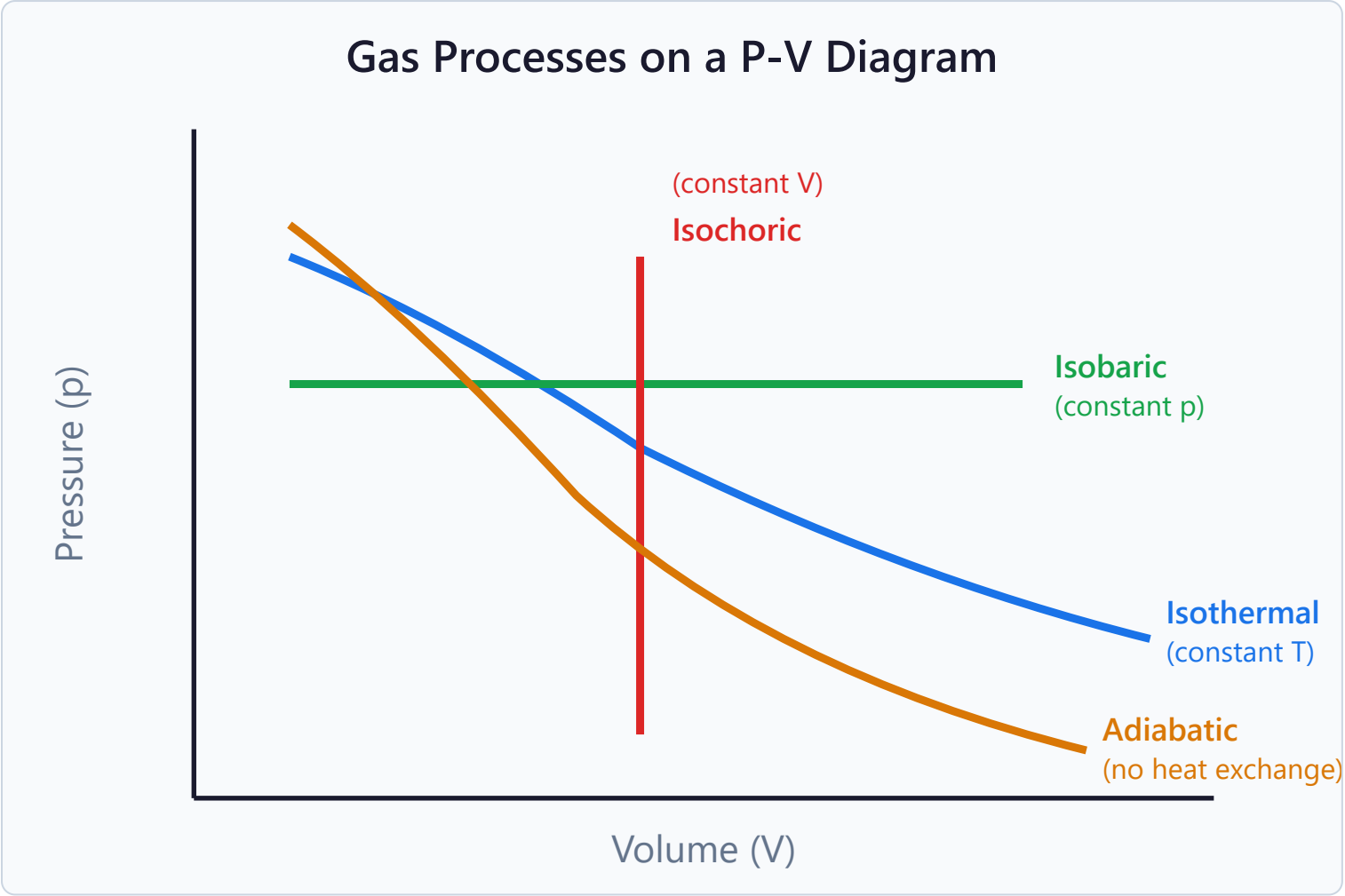
$$V_1 / T_1 = V_2 / T_2$$

Pressure Law (Gay-Lussac's Law)

Pressure law: For a fixed mass of gas at constant volume, the pressure is directly proportional to the absolute temperature.
p / T = constant (at constant V)

$$p_1 / T_1 = p_2 / T_2$$

Microscopic explanation: At constant volume, increasing temperature means molecules move faster (higher average KE). They hit the walls harder (greater impulse per collision) and more frequently, so pressure increases.



Exam Tip: All gas law calculations require temperature in **kelvin**. Convert from Celsius: T(K) = T(°C) + 273.15. A common error is forgetting to convert. A temperature of 0°C is NOT zero on the kelvin scale — it is 273 K.

Worked Example: A gas occupies 2.0×10^{-3} m^3 at 20°C and 1.0×10^5 Pa. It is heated to 100°C at constant pressure. What is its new volume?

Using Charles's law (constant pressure): $V_1/T_1 = V_2/T_2$

$T_1 = 20 + 273 = 293$ K, $T_2 = 100 + 273 = 373$ K

$V_2 = V_1 \times T_2/T_1 = 2.0 \times 10^{-3} \times 373/293 =$ **2.5×10^{-3} m^3**

9.5 Ideal Gas Equation

Combining the three gas laws gives the **ideal gas equation**:

$$pV = nRT = NkT$$

where:

Symbol	Quantity	Value / Unit
p	Pressure	Pa
V	Volume	m^3
n	Number of moles	mol
R	Molar gas constant	8.314 J mol^{-1} K^{-1}
N	Number of molecules	dimensionless
k	Boltzmann constant	1.38×10^{-23} J K^{-1}
T	Absolute temperature	K

Key relationships: $\mathbf{R = N_A \times k}$, where $N_A = 6.02 \times 10^{23}$ mol^{-1} (Avogadro's number). Also: $N = n \times N_A$.

An ideal gas is one that obeys the ideal gas equation (pV = nRT) at all pressures and temperatures. The assumptions are: (1) molecules have negligible volume, (2) no intermolecular forces except during collisions, (3) collisions are perfectly elastic. Real gases approximate ideal behaviour at **low pressures** and **high temperatures** (when molecules are far apart and intermolecular forces are negligible).

Worked Example: A sealed container of volume 0.050 m^3 holds nitrogen gas at 27°C and 2.0×10^5 Pa. Calculate: (a) the number of moles, (b) the number of molecules.

(a) pV = nRT, so n = pV/(RT) = $(2.0 \times 10^5 \times 0.050) / (8.314 \times 300)$ = 10 000 / 2494 = **4.01 mol**

(b) $N = nN_A = 4.01 \times 6.02 \times 10^{23}$ = **2.41×10^{24} molecules**

9.6 Kinetic Theory of Gases

Assumptions of the Kinetic Theory

1. The gas contains a **very large number** of molecules in **rapid, random motion**
2. The **volume of the molecules** is negligible compared to the volume of the container
3. Collisions between molecules and with the walls are **perfectly elastic** (no kinetic energy is lost)
4. The **time of a collision** is negligible compared to the time between collisions
5. There are **no intermolecular forces** between molecules (except during collisions)
6. Molecules exert pressure on the container walls due to their **collisions with the walls**

The Kinetic Theory Equation

$$pV = \tfrac{1}{3} Nm\langle c^2\rangle$$

where N is the number of molecules, m is the mass of one molecule, and $\langle c^2\rangle$ is the mean square speed.

The **root mean square (rms) speed** is: $c_{rms} = \sqrt{\langle c^2\rangle}$

Deriving the Pressure Equation

The derivation considers one molecule bouncing between two walls of a cubic container of side L:

1. A molecule of mass m with x-component of velocity c_x hits a wall
2. Change in momentum per collision: $\Delta p = 2mc_x$ (elastic collision, velocity reverses)
3. Time between successive hits on the same wall: $\Delta t = 2L/c_x$
4. Force from one molecule: $F = \Delta p/\Delta t = 2mc_x / (2L/c_x) = mc_x^2/L$
5. For N molecules: total force $= Nm\langle c_x^2\rangle/L$
6. Pressure = Force/Area $= Nm\langle c_x^2\rangle/(L \times L^2) = Nm\langle c_x^2\rangle/V$

7. Since $\langle c^2 \rangle = \langle c_x^2 \rangle + \langle c_y^2 \rangle + \langle c_z^2 \rangle = 3\langle c_x^2 \rangle$ (by symmetry): $p = Nm\langle c^2 \rangle/(3V)$

Molecular Kinetic Energy and Temperature

Combining $pV = nRT$ with $pV = \frac{1}{3}Nm\langle c^2 \rangle$:

$$\tfrac{1}{2}m\langle c^2 \rangle = \tfrac{3}{4}kT$$

This is one of the most important results in thermodynamics: the **average translational kinetic energy of a gas molecule is directly proportional to the absolute temperature**. It depends only on temperature, not on the mass of the molecule or the type of gas.

Worked Example: Calculate the rms speed of nitrogen molecules (M = 28 g/mol = 0.028 kg/mol) at 300 K.

$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{4}kT$
$\langle c^2 \rangle = 3kT/m$

Mass of one molecule: $m = M/N_A = 0.028 / 6.02 \times 10^{23} = 4.65 \times 10^{-26}$ kg
$\langle c^2 \rangle = 3 \times 1.38 \times 10^{-23} \times 300 / (4.65 \times 10^{-26}) = 2.67 \times 10^5$ m^2 s^{-2}
$c_{rms} = \sqrt{(2.67 \times 10^5)} =$ **517 m s^{-1}**

Alternatively, using $c_{rms} = \sqrt{(3RT/M)} = \sqrt{(3 \times 8.314 \times 300 / 0.028)} =$ **517 m s^{-1}**

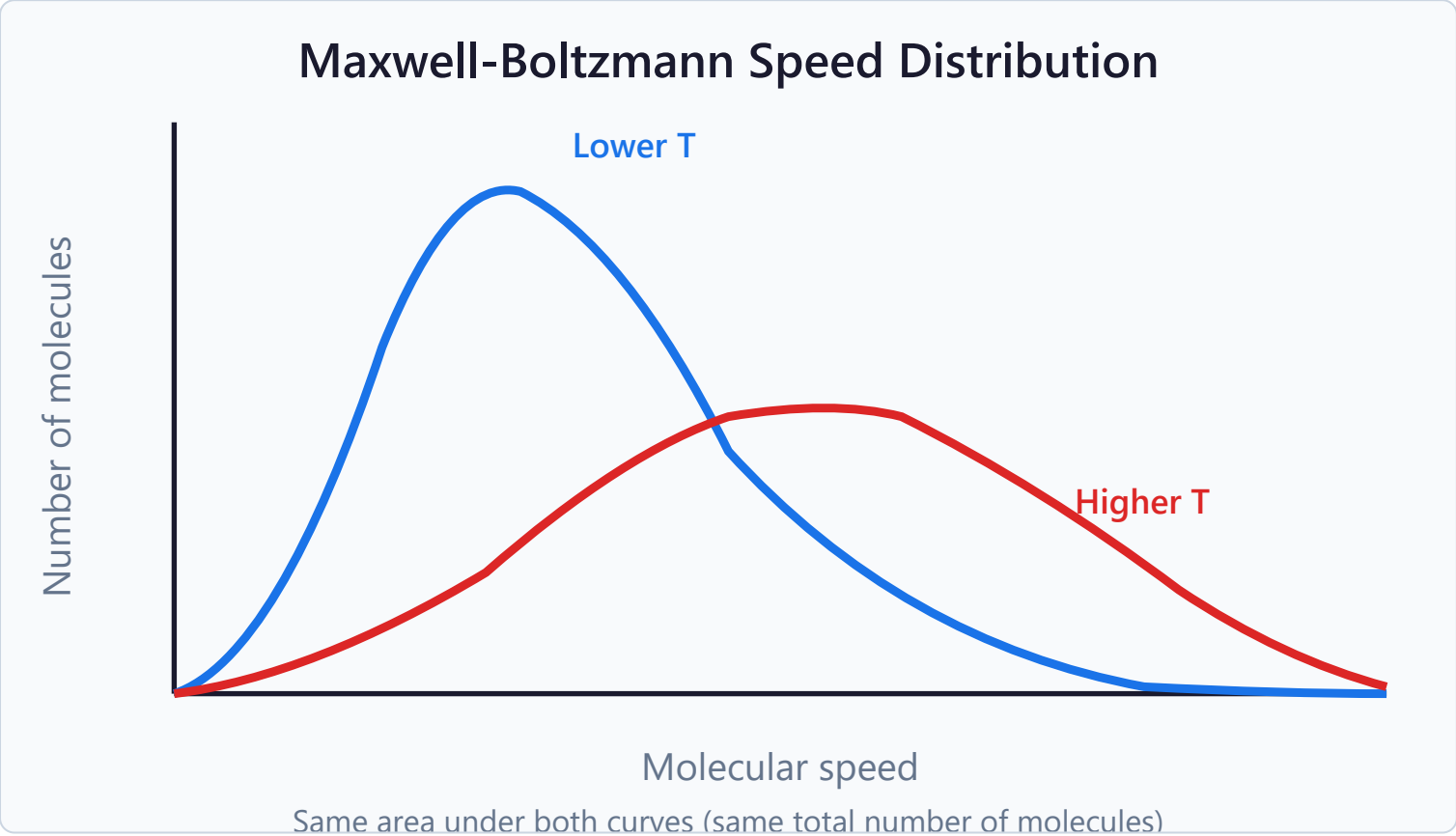
Exam Tip: The rms speed is NOT the same as the mean speed. The rms speed is always slightly larger than the mean speed because squaring the speeds gives more weight to faster molecules. In calculations, always use $c_{rms} = \sqrt{(3kT/m)}$ or $\sqrt{(3RT/M)}$.

Maxwell-Boltzmann Distribution

Not all molecules in a gas have the same speed. The distribution of molecular speeds is described by the Maxwell-Boltzmann distribution:

- The distribution is **not symmetrical** — it has a tail extending to high speeds
- The **peak** (most probable speed) shifts to higher speeds at higher temperatures

- The distribution **broadens** and **flattens** at higher temperatures
- The **area under the curve** is constant (total number of molecules does not change)
- The mean speed, rms speed, and most probable speed are all different, with: most probable < mean < rms



9.7 Absolute Zero and the Kelvin Scale

Absolute zero (0 K = −273.15°C) is the temperature at which molecules have minimum kinetic energy (they do not completely stop due to quantum zero-point energy, but classically they would have zero KE). It is the lowest possible temperature and cannot actually be reached (Third Law of Thermodynamics).

The kelvin scale is defined so that a temperature change of 1 K is equal to a temperature change of 1°C. The only difference is the zero point: 0 K = −273.15°C.

Exam Tip: In the ideal gas equation and kinetic theory, you MUST use absolute temperature (kelvin). If you use Celsius, you will get the wrong answer. A negative Celsius temperature does not mean negative kinetic energy — it just means below the freezing point of water.

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